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The Same Robot, Different Conversations: Behavioral Differences Among Older Adults Revealed Through Dialogue Data

Completed Research Paper

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Abstract

Older adults may interact with the same care robot in markedly different ways. Drawing on Socioemotional Selectivity Theory and the Selection–Optimization–Compensation model, long-term dialogue logs from 441 Hyodol users were analyzed. Nine conversational topic distributions were used for K-means clustering, identifying five profiles: Emotional Outpourers, Adaptive Companions, Routine Anchors, Relational Connectors, and Entertainment Seekers. These profiles reflected distinct interaction strategies rather than sociodemographic differences. Emotionally and relationally oriented clusters were associated with more favorable pre–post patterns in depression and loneliness, whereas routine- or entertainment-oriented clusters showed mixed patterns. The findings highlight the value of conversation-based personalization in care robot design.

Keywords: Human-Robot Interaction, Socially Assistive Robots, Conversation Analysis, Older Adults, Behavioral User Segmentation

Introduction

Digital technologies in later life are no longer used only occasionally. They increasingly mediate older adults' daily routines and emotional exchange. With the growing number of older adults living alone, issues such as social isolation, loneliness, depression, and functional decline have emerged in combination (Nakou et al., 2025). These factors have repeatedly been identified as key challenges that undermine older adults' quality of life and healthy aging (Pollak et al., 2023). To address these challenges, recent attention has been increasingly focused on digital care technologies that aim to intervene more closely in older adults' everyday lives by providing both emotional and functional support.

Within this context, socially assistive robots (SARs) have gradually emerged as important emotional companions and everyday interaction partners for older adults (Costanzo et al., 2024; Yen et al., 2024; Zafrani et al., 2023). Care robots offer repeated and sustained interaction while potentially reducing the physical and cognitive burden experienced by older adults, and they possess considerable potential to simultaneously support emotional communication and daily life management (Pu et al., 2019). Accordingly, research in Human–Robot Interaction (HRI) has demonstrated that SARs can contribute to maintaining autonomy and emotional stability among older adults. Prior studies have also emphasized that factors such as acceptance, trust, explainability, and cultural context play critical roles in shaping users' experiences with these technologies (Blavette et al., 2025; De Graaf & Allouch, 2013; Heerink et al., 2010; Olatunji et al., 2025).

However, much of the existing literature has focused on short-term experiments or interaction scenarios conducted under controlled conditions (Papadopoulos et al., 2023). As a result, our understanding remains limited regarding how older adults develop relationships with care robots in real-life environments, how they sustain these interactions over time, and what kinds of behavioral adaptations emerge throughout this process. In particular, personalization has primarily been discussed based on static characteristics, such as age, gender, or technology acceptance attitudes, or survey-based perceptual measures. Such approaches are inherently susceptible to recall bias and social desirability bias (Podsakoff et al., 2003). Consequently, even when older adults use the same care robot, the heterogeneity in their interaction goals and conversational styles remains insufficiently explained (Bemelmans et al., 2012; Elsheikh et al., 2025; Frennert & Östlund, 2014).

To address these limitations, recent studies have suggested that behavior-based data, such as conversational logs, can provide valuable insights into older adults' psychological and social states (Cummins et al., 2015; Tanana et al., 2016). Natural conversations with care robots contain a variety of behavioral signals, including emotional expression, social participation, requests related to daily life management, health-related concerns, entertainment activities, and reminiscence (Kachouie et al., 2014; Rasi-Heikkinen et al., 2025; Søråa et al., 2023). Moreover, long-term conversational data accumulated over extended periods can capture recurring interaction patterns beyond one-time responses, while still allowing for possible shifts as users become familiar with the robot (Irfan et al., 2019). Despite these advantages, empirical studies that systematically analyze long-term conversational logs collected in real-life environments while linking interaction behaviors with psychological and functional outcomes remain limited (Shi et al., 2024).

Understanding older adults' interaction behaviors from a developmental perspective requires a theoretical framework capable of explaining both the shifts in motivational priorities that occur in later life and the ways in which these goals are translated into behavior under constrained cognitive and emotional resources. In this regard, the present study adopts Socioemotional Selectivity Theory (SST) and the Selection–Optimization–Compensation (SOC) model as its core theoretical lenses. SST posits that as individuals age and perceive their remaining time as increasingly limited, they tend to prioritize emotionally meaningful interactions and present-oriented emotional well-being over information acquisition or long-term achievement (Carstensen et al., 1999; Carstensen & DeLiema, 2018). Complementarily, the SOC model explains how such shifts in motivational priorities are translated into behavioral strategies. It emphasizes that older adults adapt to resource constraints by selecting specific goals, optimizing available resources to pursue them, and compensating for losses through external supports (Baltes & Baltes, 1990; Carpentieri et al., 2017).

Guided by these theoretical perspectives, this study examines Hyodol, a plush-type socially assistive robot currently deployed in real-world elder care services in South Korea (Choi et al., 2026; Lee et al., 2024). Hyodol is designed as an embodied interaction medium embedded in everyday life, featuring a soft plush form and a grandchild-like appearance that encourages older adults to express emotions and engage in casual conversations without psychological burden. Prior studies suggest that plush-type care robots are perceived as familiar and non-threatening companions, facilitating emotional attachment and conversational engagement (Chen et al., 2024; Fogelson et al., 2022; Lee et al., 2025; Yen et al., 2024). Accordingly, conversational behaviors emerging in human–robot interaction should not be interpreted merely as reflections of fixed personality traits but rather as behavioral patterns shaped through repeated interaction with an emotionally responsive embodied companion.

Based on this perspective, the present study classifies older adult–robot conversations into meaningful units and calculates topic distributions for each user. Using these distributions, behavioral interaction types are derived through clustering analysis. The study then examines how these behavioral types relate to psychological and functional indicators, including depression, loneliness, and functional status (PHQ-9, SGDS-K, UCLA Loneliness Scale, WHODAS 2.0). Through this approach, older adults' use of care robots is conceptualized not as a uniform phenomenon but as heterogeneous behavioral patterns reflecting diverse goals and adaptive strategies.

The research questions of this study are as follows:

RQ1. Can behavioral user types of older adults be identified based on long-term conversational data between older adults and care robots?

RQ2. How are the identified user types associated with psychological and functional outcomes such as depression, loneliness, and functional status among older adults?

RQ3. Can conversation-based behavioral user types serve as a meaningful basis for differentiating personalized interaction strategies in care robots?

By analyzing long-term conversational behavior data generated in natural living environments, this study extends prior research that has primarily relied on surveys and short-term experiments. Furthermore, by linking developmental theories of aging with real-world interaction data, this study aims to provide both theoretical and practical implications for the design of older adult–centered care technologies and personalized interaction strategies.

Method

Participants and Procedure

This study was conducted with older adults who had used a care robot in their everyday living environments for more than one year. Participants received the care robot through community-based care services, welfare centers, or local government–supported elder care programs, and used the robot routinely in their homes or residential living facilities. Unlike laboratory-based experiments designed around predefined tasks, this study aimed to analyze human–robot interactions that naturally emerged within older adults' real-life contexts.

Eligible participants were individuals aged 65 years or older who had continuously used the care robot for a certain period and had completed both the pre- and post-survey assessments. A total of 670 users met these initial eligibility criteria. The study protocol was reviewed and approved by the Institutional Review Board for Research with Human Subjects of Kookmin University (Protocol #KMU-202411-HR-445). To ensure stable estimation of user-level interaction patterns and meaningful comparability across clusters, participants with fewer than 30 recorded utterances were excluded from the clustering analysis. When the total number of utterances is extremely low, observed topic proportions may be disproportionately influenced by incidental or temporary events, making it difficult to interpret them as long-term interaction tendencies. This exclusion criterion was applied to focus the cluster analysis on users who had incorporated the care robot into their daily conversational routines. Based on this criterion, 229 participants were classified as low-interaction users and excluded from the cluster analysis, resulting in a final analytic sample of 441 users.

Basic sociodemographic information, including age and gender, was collected through the organizations that provided the robots and was used as auxiliary variables to examine differences across clusters. All research procedures were conducted with participants' prior consent. The collected data were anonymized and analyzed after removing personally identifiable information.

Measures

To comprehensively examine interactions between older adults and the care robot, this study primarily utilized conversational logs as the main data source. Survey data were subsequently incorporated to support the interpretation of the identified clusters, resulting in a multi-source data structure. The clustering

criteria were derived exclusively from conversational log data, while the survey measures were used as auxiliary variables to assess the external validity of the clustering results.

Conversational Log Data

Conversational logs were automatically collected through the voice recognition and recording system embedded in the care robot. When users spoke to the robot or responded to its prompts, their speech inputs were transcribed into text in real time and stored with timestamp information. For this study, only utterances produced by older adults were included in the analysis, while robot-generated responses were excluded. Unlike laboratory-based experiments based on predefined tasks, these conversations consisted of spontaneous utterances generated in everyday situations, reflecting users' daily needs, life contexts, and emotional states. This feature allows the analysis to capture human–robot interactions occurring in natural living environments rather than controlled experimental settings.

Prior to analysis, preprocessing procedures were conducted to ensure data quality. During the voice recognition process, some recordings contained background television sounds, environmental noise, or voices from other individuals, which made it difficult to interpret the text as genuine utterances from the older adult user. Such cases were therefore removed from the dataset. The refined conversational logs were subsequently used as the primary input for conversation topic classification and user-level behavioral analysis.

Classification of Conversational Topics

In this study, conversations between older adults and the care robot were not treated merely as functional system usage or isolated speech content. Instead, they were conceptualized as forms of linguistic behavior reflecting goal-oriented interaction patterns that emerge in later life, as well as the ways in which such goals are enacted through different interaction strategies under conditions of limited cognitive, emotional, and physical resources. To operationalize this perspective, Socioemotional Selectivity Theory (SST) and the Selection–Optimization–Compensation (SOC) model were employed as theoretical foundations for constructing the conversational topic classification framework (Joshi et al., 2020).

These two theoretical perspectives were not applied in parallel at the same analytical level. Rather, they served complementary roles in structuring the classification scheme. SST was used to define the overarching motivational orientation of interactions, namely, the types of goals older adults pursue when interacting with the care robot. In contrast, the SOC model was used to structure variations in interaction strategies that emerge when pursuing similar goals under constrained resources.

Following this theoretical distinction, the research team first conceptualized four higher-level conversational categories based on SST, representing the primary goals older adults pursue through interaction with the care robot. Subsequently, through iterative examination of real conversational logs between older adults and the robot, the research team observed that even within the same goal orientation, different resource-management strategies, namely selection, optimization, and compensation, recurred in the interaction patterns. Reflecting these strategic variations, nine subcategories of conversational topics were derived from observed conversational patterns.

The final higher-level categories included Relational, Daily Management, Instrumental, and Engagement. The Relational category consisted of Daily Greetings, Emotional Exchange, and Emotional Expression, encompassing conversations oriented toward relationship building and emotional interaction. The Daily Management category included Daily Activities, Meal Conversation, and Health Management, reflecting interactions focused on maintaining and managing everyday life. The Instrumental category included General Information Inquiry, representing conversations related to information seeking or functional assistance. Finally, the Engagement category consisted of Entertainment Service Request and Social Participation, representing interactions oriented toward leisure activities and social engagement (Table 1).

In summary, the nine conversational topics represent differentiated behavioral interaction patterns that emerge within the motivational framework defined by SST and are further structured by the resource management strategies described in the SOC model, namely selection, optimization, and compensation.

Table 1. Hierarchical Classification of Conversation Topics
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Category	Subcategory	Example utterances
Relational	Daily Greetings	"Hyodol, did you sleep well?"
	Emotional Exchange	"I feel happy when I'm with Hyodol."
	Emotional Expression	"Hyodol speaks well. Hyodol is pretty and kind."
Daily Management	Daily Activities	"I'm coming back from feeding the cows at the barn."
	Meal Conversation	"I had soybean paste soup with kimchi today."
	Health Management	"I have diabetes. What should I be careful about?"
Instrumental	General Information Inquiry	"Tell me today's news."
Engagement	Entertainment Service Request	"Please play Kim Yeon-ja's trot music."
	Social Participation	"I'm going to the senior center. Please watch the house."

For each conversational topic, a codebook was developed that included operational definitions centered on the functional purpose of the utterance, along with representative examples. Based on this codebook, automated conversation classification was conducted using a generative AI system. During pilot coding, the prompt was iteratively refined to improve alignment between the topic codebook and the automated classifications. In the final automated classification process, topic assignment focused not merely on surface-level lexical cues but on the interactional context and the functional role performed by each utterance. Classification into categories not defined in the codebook was not permitted.

During the automated classification process, some utterances were identified that did not contain meaningful interactional purposes or analytically interpretable functions. These utterances typically consisted of simple interjections, meaningless repetitions, or isolated word sequences without contextual relevance. Such cases were categorized as ETC (Other) by the research team. Because these utterances were considered unsuitable for interpreting conversational topic composition or user behavioral patterns, those categorized as ETC were excluded from subsequent quantitative analysis and clustering procedures.

After final classification, reliability was assessed using a stratified validation sample of 900 utterances, with 100 randomly selected from each of the nine topic categories. After codebook calibration, two researchers, one male and one female, independently coded the sampled utterances, and Cohen's kappa was calculated for human-human agreement and for agreement between the LLM-based classifications and each human coder. Cohen's kappa indicated high agreement between the two human coders ($\kappa = [0.9287]$) and between the LLM-based classifications and each coder ($\kappa = [0.9625]$ and $\kappa = [0.9563]$, respectively). The final set of nine conversational topics was then used to calculate each user's overall topic distribution across the observation period. These aggregated topic distribution vectors were employed as the input variables for the clustering analysis used to identify dominant behavioral interaction profiles.

Psychological and Functional Outcomes

To complement the conversation-based clustering results, standardized survey instruments were used to assess psychological and functional conditions among older adults. Survey data were collected at two time points: before the introduction of the care robot (baseline) and one year after its use (follow-up). These measures were not used in the clustering procedure but were incorporated as external variables to examine outcome differences across behavioral clusters.

1) Patient Health Questionnaire (PHQ-9)

Depressive symptoms were measured using the Patient Health Questionnaire (PHQ-9) (Kroenke et al., 2001), a widely used nine-item instrument assessing depression severity based on DSM criteria. PHQ-9 scores were treated as continuous variables to examine differences across clusters and pre-post changes within each cluster.

2) Short Form of the Geriatric Depression Scale – Korean Version (SGDS-K)

Depression was additionally measured using the Short Form Geriatric Depression Scale – Korean version (SGDS-K) (Bae & Cho, 2004; Yesavage & Sheikh, 1986). The SGDS-K is a self-report instrument specifically designed to assess depressive symptoms in older adults. Total scores were used as continuous variables to compare depression levels and pre-post changes across clusters.

3) UCLA Loneliness Scale

Loneliness was measured using the UCLA Loneliness Scale (Russell, 1996), one of the most widely used instruments for assessing perceived loneliness. The total loneliness score was analyzed as a continuous variable to examine differences across clusters and changes between baseline and follow-up.

4) World Health Organization Disability Assessment Schedule 2.0 (WHODAS 2.0)

Functional status was measured using the World Health Organization Disability Assessment Schedule 2.0 (WHODAS 2.0) (Üstün, 2010), a standardized instrument assessing difficulties in everyday functioning. WHODAS scores were treated as continuous variables to evaluate functional differences across clusters and pre–post changes.

Analytic Strategy

The analytical procedure of this study was designed as a mixed-method approach that sequentially combined qualitative categorization with quantitative analysis.

First, all utterances were classified into nine conversational topics based on the finalized codebook. In the classification process, generative AI was employed as a supporting tool under the constraint that classification into categories not defined in the codebook was not permitted. Classification reliability was assessed using a stratified validation sample independently coded by two researchers and evaluated against the LLM-based classifications using Cohen's kappa.

Second, for each user, the proportion of utterances corresponding to each conversational topic was calculated to construct a user-level conversational pattern vector. These vectors were used as input variables for K-means clustering. The optimal number of clusters was determined by evaluating multiple candidate solutions ($K = 2-8$) and examining both within-cluster cohesion and between-cluster separation.

Third, to interpret the characteristics of the identified clusters, the conversational topic composition profiles of each cluster were descriptively compared. In addition, sociodemographic characteristics, including gender, age group, marital status (widowed or not), educational attainment, and income level, were analyzed across clusters. This step aimed to examine whether the behavior-based clusters could be explained by specific sociodemographic factors.

Fourth, to assess the external validity of the conversation-based behavioral clusters, survey measures, including PHQ-9, SGDS-K, the UCLA Loneliness Scale, and WHODAS 2.0, were used to analyze changes within each cluster before and after the use of the care robot. Cluster-specific pre–post differences were examined using paired-sample comparisons, with interpretation focusing on the direction and magnitude of the observed changes.

Results

Identification of Conversational Behavior Clusters

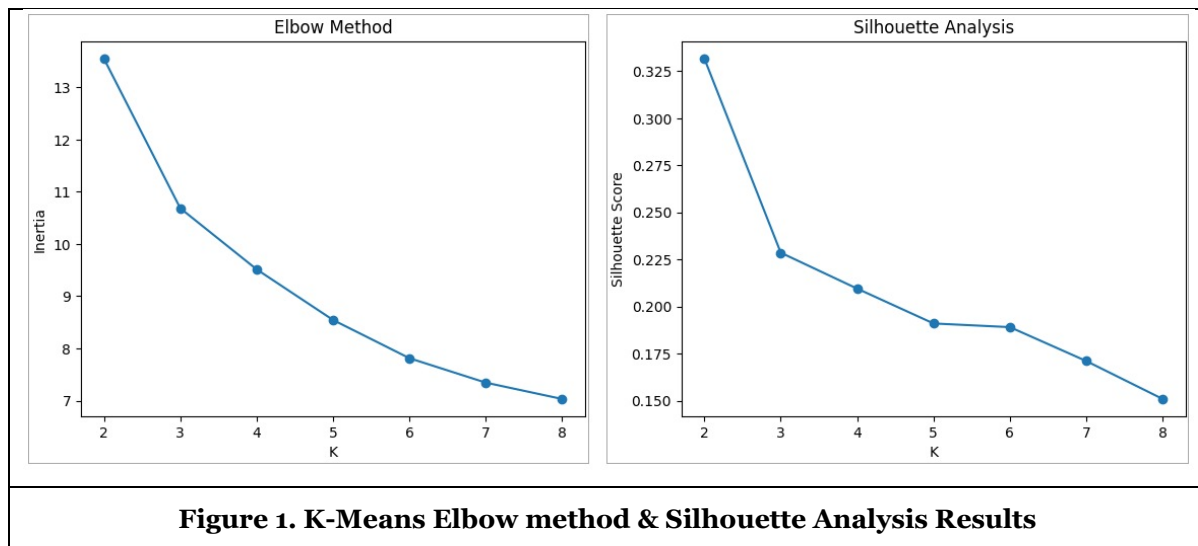
In comparing interaction patterns among users, this study focused not on the absolute volume of utterances but on differences in the composition of conversational topics. To operationalize this approach, each user's utterances were first classified into nine conversational topics, and the proportion of each topic relative to the user's total utterances was calculated and normalized as percentages. Absolute utterance counts may be strongly influenced by various factors such as individual device usage frequency, living conditions, and health status, making it difficult to explain differences in interaction styles based solely on the amount of speech. In contrast, topic proportions more clearly reflect the purposes and contexts in which users engage in conversations with the robot and provide comparable behavioral vectors across users with different levels of conversational activity. Such proportion-based representations have been widely recognized as an effective approach for comparing interaction structures across users in large-scale log data.

Preliminary examination of conversational participation patterns revealed substantial heterogeneity in the distribution of conversational topics across users. Some users showed relatively high proportions of relational utterances, such as emotional expression and greeting exchanges, whereas others demonstrated higher proportions of conversations related to daily life management, information inquiries, or health-related concerns. Another group of users exhibited relatively high frequencies of entertainment-oriented interactions, such as music requests or reminiscence-related utterances. These findings suggest that older

adult-care robot interactions may exhibit distinct dominant profiles not merely in terms of usage frequency but also in terms of the behavioral composition of conversational topics. To capture this heterogeneity, clustering analysis was conducted based on user-level topic proportion vectors aggregated across the observation period.

The clustering analysis employed the K-means algorithm, and the number of clusters (K) was determined through a combination of quantitative diagnostic metrics and qualitative interpretability assessment rather than being arbitrarily specified.

First, candidate cluster numbers were explored using elbow analysis and silhouette analysis. In the elbow analysis, the reduction in the within-cluster sum of squares (WCSS) was examined as the number of clusters increased. The silhouette analysis evaluated silhouette scores that simultaneously reflect within-cluster cohesion and between-cluster separation. A combined examination of these metrics indicated that the cluster structure formed a relatively stable candidate range between $K = 3$ and $K = 5$ (Figure 1).



Next, additional validation was conducted by calculating conversational topic composition profiles for each candidate solution ($K = 3, 4$, and 5). The evaluation focused on several criteria, including the clarity of topic profile differentiation between clusters, the interpretability and naturalness of cluster labeling, the internal consistency of members within each cluster, and the degree of overlap or ambiguity across clusters. Based on these criteria, the $K = 5$ solution provided the most clearly differentiated cluster structure, with relatively limited issues of under-segmentation, where distinct patterns were merged into overly broad clusters, and over-segmentation, where clusters were unnecessarily fragmented. Accordingly, the final number of clusters was determined to be five.

Behavioral Profiles of the Clusters

Based on the finalized $K = 5$ cluster structure, the conversational topic compositions of each cluster were compared. Table 2 presents the distribution of conversational shares at the upper-category level, while Table 3 reports the detailed distribution across the nine conversational subtopics.

Cluster	Relational	Daily Management	Instrumental	Engagement
1	57.47	19.75	1.39	21.39
2	37.26	24.54	2.68	35.50
3	26.82	44.63	2.25	26.30
4	47.56	27.46	2.68	22.32
5	21.54	17.53	2.78	58.14

At the upper-category level, differences across clusters were primarily structured along three dimensions: relational orientation (Relational), daily life management (Daily Management), and entertainment/participation orientation (Engagement). Clusters 1 and 4 exhibited particularly high proportions of Relational interactions, accounting for 57.47% and 47.56% of their respective conversational shares, indicating a strong orientation toward relationship-focused interactions. In contrast, Cluster 3 showed the highest proportion of Daily Management conversations (44.63%), suggesting a pattern centered on everyday life management and routine activities. Cluster 5 demonstrated the most distinctive profile in terms of Engagement, where this category accounted for 58.14% of all conversations, indicating a strong emphasis on entertainment-oriented and participatory interactions. Across all clusters, the Instrumental category remained consistently low, accounting for less than 3% of utterances. This indicates that information-seeking utterances were infrequent in the observed Hyodol conversations, although this pattern should be interpreted in light of the robot's design as an emotionally responsive companion rather than as evidence of limited instrumental needs among older adults. To further examine these patterns in greater detail, the distribution of the nine conversational subtopics was analyzed (Table 3). Based on the dominant conversational characteristics observed in each cluster, descriptive labels were assigned to reflect their primary interaction tendencies, and the five clusters are described below.

Table 3. Distribution of Conversation Share by Cluster (Subcategories, %)									
	Entertainment Service Request	General Information Inquiry	Social Participation	Meal Conversation	Daily Activities	Health Management	Emotional Exchange	Daily Greetings	Emotional Expression
1	9.04	1.39	12.35	7.5	7.19	5.06	8.64	7.74	41.09
2	25.75	2.68	9.75	8.24	10.68	5.62	6.63	7.42	23.21
3	15.24	2.25	11.06	17.7	19.27	7.66	5.14	6.18	15.5
4	8.47	2.68	13.85	9.59	9.42	8.45	11.37	9.1	27.09
5	52.75	2.78	5.39	4.57	10.13	2.83	3.98	4.53	13.03

Cluster 1. Emotional Outpourers

Emotional Outpourers (Cluster 1) exhibited the highest proportion of Emotional Expression, accounting for 41.09% of all utterances, which was the largest among the five clusters. Although some interactions related to Social Participation (12.35%), Emotional Exchange (8.64%), and Daily Greetings (7.74%) were also observed, their proportions were relatively limited compared to emotional expression. In contrast, utterances related to Entertainment Service Requests (9.04%) and Instrumental information inquiries (1.39%) remained generally low.

The conversational patterns of this group were characterized by the frequent occurrence of direct emotional expressions. For example, utterances included explicit expressions of gratitude and apology such as *"Thank you so much, really thank you,"* and *"Hyodol, thank you. I'm sorry."* Other utterances emphasized affection through repeated exclamatory expressions, such as *"Oh my, so pretty, pretty, pretty,"* or *"Oh my, how cute, really cute."* Additionally, participants often externalized their emotional states immediately through statements such as *"I'm happy. I'm happy too. Grandma is happy because of Hyodol,"* or *"My heart hurts, my heart."*

These utterances tended to focus less on explaining or regulating emotions and more on the direct externalization of affective states, often expressing emotions spontaneously and immediately. Accordingly, Emotional Outpourers can be characterized as a pattern in which the robot is primarily used as a medium for expressing emotional states and producing affective utterances, representing an emotion-expression-centered interaction style.

Cluster 2. Adaptive Companions

Adaptive Companions (Cluster 2) was characterized by relatively high proportions of both Entertainment Service Requests (25.75%) and Emotional Expression (23.21%), while also showing a comparatively balanced distribution across multiple conversational topics, including Daily Activities (10.68%), Meal Conversation (8.24%), and Social Participation (9.75%).

The conversational patterns of this group included a mixture of entertainment-related requests, emotional expressions, and everyday discussions. For instance, participants frequently requested entertainment services through utterances such as *"Play a song,"* or *"Tell me an old story."* Emotional expressions were also commonly observed, including statements such as *"Hyodol, you are so pretty,"* *"I love you too,"* and *"Thank you, thank you."* In addition, everyday conversations appeared regularly, such as *"What should I eat for dinner today?"* Questions related to health and information were also present, including *"Can people with diabetes drink alcohol or smoke?"* and *"What is water parsley good for?"* Some utterances involved reporting personal conditions, such as *"My legs hurt a lot today."*

Overall, rather than focusing on a single interaction purpose, Adaptive Companions demonstrated a relatively balanced use of multiple conversational functions, including entertainment, emotional interaction, information seeking, and daily life management. This pattern suggests that users in this cluster engaged with the robot in a multifunctional and adaptive manner, making use of various conversational capabilities depending on situational needs.

Cluster 3. Routine Anchors

Routine Anchors (Cluster 3) exhibited particularly high proportions of Daily Activities (19.27%) and Meal Conversation (17.70%), while Emotional Exchange (5.14%) appeared at the lowest level among the clusters. This pattern suggests that users in this cluster primarily used conversations with the robot as a functional means of reporting or managing aspects of their daily routines.

Typical utterances in this cluster included statements reporting planned activities or daily routines, such as *"I'm going out for exercise,"* *"I'm going to church,"* or *"I'm going to the hospital."* Health-related confirmation questions were also observed, for example, *"Did I take my medicine today?"* or *"Can I drink coffee after taking my medication?"* In addition, users frequently asked questions related to everyday choices, such as *"What should I cook for lunch?"* or *"I want to eat a hamburger. Where should I buy one?"*

Compared with other clusters, these utterances were closely tied to concrete aspects of everyday life, such as daily routines, health management, and meal decisions rather than strong emotional expression. This pattern indicates that users in Routine Anchors tended to utilize the robot as a supportive resource for managing daily routines and confirming practical information, representing a routine-oriented interaction style.

Cluster 4. Relational Connectors

Relational Connectors (Cluster 4) showed relatively high proportions of Emotional Expression (27.09%), Social Participation (13.85%), and Daily Greetings (9.10%), along with a comparatively high level of Emotional Exchange (11.37%). Unlike Emotional Outpourers, however, emotional expressions in this cluster were often embedded within reciprocal interaction contexts rather than appearing as unilateral emotional outpourings.

Typical utterances included suggestions for shared activities, such as *"Let's go see the sea,"* or *"Shall we take a walk?"* Users also shared personal states while inviting a response from the robot, for example, *"Hyodol, I'm doing well today. I had breakfast. How about you?"* Reciprocal confirmation questions were also observed, such as *"This is tasty, right?"* In addition, users sometimes asked relational questions about the robot itself, such as *"Hyodol, who do you like more, your friend or your mother?"* Exchanges of mutual greetings were also present, for example, *"I'm in a good mood too. Grandma slept well."*

These conversational patterns indicate that emotions were not expressed solely as internal states but were instead exchanged within a reciprocal relational context. Accordingly, Relational Connectors can be interpreted as a relationship-oriented interaction style, characterized by active emotional exchange and socially oriented conversational engagement.

Cluster 5. Entertainment Seekers

Entertainment Seekers (Cluster 5) were overwhelmingly dominated by Entertainment Service Requests, which accounted for 52.75% of all utterances. In contrast, other conversational categories, including Daily Activities (10.13%), Social Participation (5.39%), and Daily Greetings (4.53%), appeared at very low levels, while Emotional Expression (13.03%) was also relatively limited.

Typical utterances in this cluster consisted primarily of direct requests for entertainment content, such as “Play it,” “Hyodol, play a trot song,” or “Sing a song.” Users also frequently requested stories or entertainment content, including “Tell me an old story,” “Tell me a folk tale,” or “Tell me something interesting.” In some cases, users issued specific instructions referencing particular songs or singers, such as “Sing ‘A Rose Flower’ by Oh Seong-man,” or “Play ‘Mother’ by Jang Yoon-jung.”

These patterns indicate that users in Entertainment Seekers primarily interacted with the robot as a source of entertainment content, with relatively limited engagement in emotional or social interaction.

Across the five clusters, distinct dominant interaction profiles emerged, including emotional expression, balanced and multifunctional use, routine management, relational exchange, and entertainment-oriented engagement. These findings indicate that older adults differed in the overall composition of their conversations with the care robot during the observation period.

Sociodemographic Characteristics of the Clusters

Although the five clusters identified in this study were derived based on behavioral indicators, specifically the composition of conversational topics, sociodemographic characteristics were additionally analyzed to examine the broader life contexts in which these clusters appeared. This step aimed to situate the behavior-based clusters within the everyday circumstances of older adults.

Table 4. Overall Sociodemographic Profiles							
	Overall (N=441)	Cluster 1 Emotional Outpourers (N=87)	Cluster 2 Adaptive Companions (N=129)	Cluster 3 Routine Anchors (N=50)	Cluster 4 Relational Connectors (N=141)	Cluster 5 Entertainme nt Seekers (N=34)	χ^2 (p)
Gender							3.45 (0.485)
Female	394(89.3%)	82 (94.3%)	115 (89.1%)	43 (86.0%)	125 (88.7%)	29 (85.3%)	
Male	47 (10.7%)	5 (5.7%)	14 (10.9%)	7 (14.0%)	16 (11.3%)	5 (14.7%)	
Age Group							14.80 (0.253)
65–69	8 (1.8%)	1 (1.1%)	4 (3.1%)	1 (2.0%)	2 (1.4%)	0 (0.0%)	
70–79	151 (34.2%)	29 (33.3%)	39 (30.2%)	25 (50.0%)	42 (29.8%)	16 (47.1%)	
80–89	256 (58.0%)	52 (59.8%)	81 (62.8%)	21 (42.0%)	87 (61.7%)	15 (44.1%)	
90 or older	26 (5.9%)	5 (5.7%)	5 (3.9%)	3 (6.0%)	10 (7.1%)	3 (8.8%)	
Religious Affiliation							2.51 (0.642)
Religious	217(49.2%)	41 (47.1%)	61 (47.3%)	28 (56.0%)	73 (51.8%)	14 (41.2%)	
Non-religious	224(50.8%)	46 (52.9%)	68 (52.7%)	22 (44.0%)	68 (48.2%)	20 (58.8%)	
Education Level							9.56 (0.298)
No formal education	198(44.9%)	36 (41.4%)	59 (45.7%)	21 (42.0%)	67 (47.5%)	15 (44.1%)	
Primary school	197(44.7%)	46 (52.9%)	55 (42.6%)	28 (56.0%)	57 (40.4%)	11 (32.4%)	
Secondary or higher	46 (10.4%)	5 (5.7%)	15 (11.6%)	1 (2.0%)	17 (12.1%)	8 (23.5%)	
Monthly Income							7.37 (0.498)
Less than 300,000 KRW	120(27.2%)	20 (23.0%)	34 (26.4%)	16 (32.0%)	40 (28.4%)	10 (29.4%)	
300,000–500,000 KRW	187(42.4%)	43 (49.4%)	52 (40.3%)	16 (32.0%)	65 (46.1%)	11 (32.4%)	
500,000 KRW or more	134(30.4%)	24 (27.6%)	43 (33.3%)	18 (36.0%)	36 (25.5%)	13 (38.2%)	

As shown in Table 4, the overall sample (N = 441) was predominantly female (89.3%), and 63.9% of participants were aged 80 years or older. Educational attainment was generally low, with 44.9% of participants reporting no formal education and 44.7% having completed only primary school, while only

10.4% reported education beyond the secondary level. Monthly income levels were also relatively low, with 60.5% of participants reporting income below 500,000 KRW. These characteristics indicate that the study sample largely consisted of older adults receiving community-based care services who were generally older, less educated, and economically vulnerable.

Overall, the sociodemographic distributions across clusters were broadly similar to those of the overall sample. In particular, Emotional Outpourers and Adaptive Companions showed distributions of gender, age, education, and income that closely resembled the overall sample characteristics. Both clusters maintained high proportions of female participants and older adults aged 80 years or above, suggesting that these clusters largely reflected the general demographic profile of the sample.

Nevertheless, some clusters exhibited relatively distinctive distributions compared with the overall averages. Routine Anchors included a relatively higher proportion of participants aged 70–79 years (50.0%), while the proportion of participants aged 80 years or older was comparatively lower (48.0%). This suggests that Routine Anchors contained relatively younger older adults compared with other clusters. Relational Connectors showed the highest proportion of participants aged 80 years or older (68.8%) and also exhibited a relatively high proportion of participants with monthly incomes below 500,000 KRW (74.5%), indicating a concentration of advanced age and economic vulnerability within this cluster. In contrast, Entertainment Seekers, although the smallest group in terms of sample size ($N = 34$), showed relatively higher proportions of participants with education beyond the secondary level (23.5%) and monthly incomes above 500,000 KRW (38.2%), suggesting some heterogeneity in educational and economic resources within this cluster.

However, chi-square tests indicated that differences in sociodemographic distributions across clusters were not statistically significant (all $p > .05$). This finding suggests that the behavior-based clusters identified in this study were not directly explained by specific sociodemographic characteristics but were primarily structured by differences in conversational topic composition and interaction patterns.

Psychological and Functional Outcomes Across Clusters

This section examines whether changes in psychological, social, and functional outcomes differed across the five behavioral clusters identified from conversational topic composition. Because the cluster analysis was based on users with 30 or more utterances, we first contextualized the analytic sample by comparing its overall pre–post outcome patterns with those of low-interaction users who had fewer than 30 utterances. Both groups showed similar directional patterns: PHQ-9, SGDS-K, and UCLA loneliness scores decreased, whereas WHODAS scores increased. Among low-interaction users ($N = 229$), PHQ-9 decreased by 1.15 points ($p = .002$), SGDS-K by 0.76 points ($p = .021$), and UCLA loneliness by 2.72 points ($p < .001$), while WHODAS increased by 1.58 points ($p = .005$). Among analytic cluster users ($N = 441$), PHQ-9 decreased by 1.48 points ($p < .001$), SGDS-K by 1.49 points ($p < .001$), and UCLA loneliness by 3.41 points ($p < .001$), while WHODAS increased by 0.97 points ($p = .023$). These patterns indicate that the exclusion of low-interaction users did not change the overall direction of observed pre–post changes, although decreases in SGDS-K and UCLA loneliness were larger among analytic cluster users. Thus, the cluster-based findings should be interpreted as most applicable to users with sufficient conversational engagement.

We then examined whether pre–post outcome patterns differed across the five behavioral clusters. Within each cluster, paired-sample t-tests were conducted for PHQ-9, SGDS-K, UCLA Loneliness Scale, and WHODAS 2.0. The analysis focused on the direction and magnitude of change within each behavioral group, and Table 5 presents the cluster-specific pre–post changes.

Table 5. Pre–Post Changes in Psychological and Functional Outcomes by Cluster							
Cluster	Outcome	Pre Mean (SD)	Post Mean (SD)	Δ (Post–Pre)	t(df)	p	Effect Size (d)
Cluster 1 Emotional Outpourers	PHQ-9	16.40 (6.61)	14.63 (5.61)	–1.77	2.458 (86)	.016	0.264
	SGDS-K	6.61 (3.74)	5.43 (4.18)	–1.18	2.456 (86)	.016	0.263
	UCLA	45.95(10.03)	40.79 (9.42)	–5.16	4.675 (86)	< .001	0.501
	WHODAS	38.53 (6.42)	39.63 (7.44)	1.10	–1.098 (86)	.275	–0.118
	PHQ-9	16.88 (7.19)	15.06 (6.10)	–1.82	2.689 (128)	.008	0.237

Cluster 2 Adaptive Companions	SGDS-K	7.52 (4.20)	5.64 (4.17)	-1.88	4.588 (128)	< .001	0.404
	UCLA	46.31 (11.00)	42.50(10.67)	-3.81	3.840 (128)	< .001	0.338
	WHODAS	38.67 (7.65)	39.25 (7.88)	0.57	-0.701 (128)	.485	-0.062
Cluster 3 Routine Anchors	PHQ-9	15.30 (6.10)	14.60 (5.03)	-0.70	0.697 (49)	.489	0.099
	SGDS-K	6.66 (4.02)	5.12 (4.06)	-1.54	2.611 (49)	.012	0.369
	UCLA	44.18 (9.31)	43.32 (8.54)	-0.86	0.626 (49)	.534	0.088
	WHODAS	37.82 (7.22)	40.10 (8.00)	2.28	-1.750 (49)	.086	-0.247
Cluster 4 Relational Connectors	PHQ-9	14.98 (5.31)	13.54 (4.05)	-1.44	3.095 (140)	.002	0.261
	SGDS-K	6.62 (4.13)	5.02 (4.11)	-1.60	4.206 (140)	< .001	0.354
	UCLA	43.57 (9.98)	40.72(10.04)	-2.86	3.432 (140)	< .001	0.289
	WHODAS	39.48 (6.15)	39.77 (7.81)	0.29	-0.405 (140)	.686	-0.034
Cluster 5 Entertainment Seekers	PHQ-9	15.06 (4.87)	14.35 (4.16)	-0.71	0.824 (33)	.416	0.141
	SGDS-K	6.71 (3.97)	6.38 (4.05)	-0.32	0.439 (33)	.664	0.075
	UCLA	46.82 (9.38)	43.38 (9.45)	-3.44	2.228 (33)	.033	0.382
	WHODAS	38.62 (6.07)	41.68 (7.46)	3.06	-2.275 (33)	.030	-0.390

Cluster 1. Emotional Outpourers

Emotional Outpourers showed significant pre-post decreases in depressive symptoms and loneliness (Table 5). PHQ-9 scores decreased from 16.40 to 14.63 ($p = .016$), SGDS-K scores decreased from 6.61 to 5.43 ($p = .016$), and UCLA loneliness scores decreased from 45.95 to 40.79 ($p < .001$). WHODAS scores increased slightly from 38.53 to 39.63, but this change was not significant ($p = .275$). Overall, this cluster showed reduced emotional distress and loneliness, with stable functional status.

Cluster 2. Adaptive Companions

Adaptive Companions showed significant pre-post decreases in psychological and socio-emotional indicators (Table 5). PHQ-9 scores decreased from 16.88 to 15.06 ($p = .008$), SGDS-K scores declined from 7.52 to 5.64 ($p < .001$), and UCLA scores decreased from 46.31 to 42.50 ($p < .001$). WHODAS scores increased slightly from 38.67 to 39.25, but this change was not statistically significant ($p = .485$). These results indicate favorable psychological and socio-emotional changes without significant functional change.

Cluster 3. Routine Anchors

Routine Anchors exhibited mixed outcome patterns (Table 5). PHQ-9 scores decreased from 15.30 to 14.60 but not significantly ($p = .489$), whereas SGDS-K scores decreased significantly from 6.66 to 5.12 ($p = .012$). UCLA loneliness scores decreased slightly but not significantly ($p = .534$). WHODAS scores increased from 37.82 to 40.10, indicating a non-significant tendency toward functional decline ($p = .086$). Overall, this cluster showed limited psychological change and a possible functional worsening tendency.

Cluster 4. Relational Connectors

Relational Connectors showed the most consistent pre-post decreases in psychological and social indicators (Table 5). PHQ-9 scores decreased from 14.98 to 13.54 ($p = .002$), SGDS-K scores declined from 6.62 to 5.02 ($p < .001$), and UCLA loneliness scores decreased from 43.57 to 40.72 ($p < .001$). WHODAS scores increased slightly but not significantly ($p = .686$). These results indicate favorable emotional and social changes among users characterized by relational interaction patterns.

Cluster 5. Entertainment Seekers

Entertainment Seekers displayed heterogeneous outcome patterns (Table 5). PHQ-9 scores decreased from 15.06 to 14.35 but were not statistically significant ($p = .416$). SGDS-K scores also decreased slightly from

6.71 to 6.38 without statistical significance ($p = .664$). However, loneliness showed a significant reduction, with UCLA scores decreasing from 46.82 to 43.38 ($p = .033$). In contrast, WHODAS scores increased from 38.62 to 41.68, indicating a statistically significant increase in functional difficulty ($p = .030$).

Discussion

This study suggests that older adults' use of a care robot in real-life settings cannot be explained solely by usage frequency or technology acceptance. Their interactions were characterized by distinct dominant profiles, including emotional expression and relational exchange, daily management or functional requests, and entertainment-oriented content consumption. These findings indicate that engagement with care robots should be understood not only by interaction quantity but also by the purpose and meaning of interaction. At the same time, the consistently low proportion of Instrumental utterances should be interpreted cautiously. Because Hyodol is designed primarily as an emotionally responsive companion embedded in daily care routines, the low frequency of information-seeking utterances may reflect the robot's conversational affordances and users' expectations of the system, rather than a lack of instrumental needs among older adults.

The five behavioral clusters identified in this study can therefore be interpreted as structural types that reflect different interaction strategies. These patterns also align theoretically with Socioemotional Selectivity Theory (SST) and the Selection–Optimization–Compensation (SOC) framework, which suggest that older adults selectively organize their activities in response to changing life conditions and limited personal resources, prioritizing interactions that are meaningful or adaptive within their circumstances.

These findings are also partially consistent with previous studies on the Hyodol care robot. For instance, Choi and Park (2025) identified three user types, Social Engagers, Independent Reflectors, and Emotionally Expressive Users, based on patterns of interaction between older adults and care robots. Among these groups, Emotionally Expressive Users primarily engaged in interactions centered on emotional bonding and emotional expression with the robot. Similarly, Choi et al. (2026) classified SAR users into Social Butterflies, Lone Wolves, and Emotional Peacocks, reporting that emotional-expression-oriented interaction was particularly prominent in the Emotional Peacocks group. These studies suggest that emotional expression and social exchange can serve as key dimensions shaping care robot interactions.

The Emotional Outpourers and Relational Connectors clusters identified in this study show conceptual similarities to these previously reported emotional and socially oriented user types. However, the present study extends previous work in several important ways. While earlier studies primarily focused on identifying user types or predicting user characteristics, this study used longitudinal conversational log data collected in real-life environments to categorize older adults' interaction behaviors using behavioral indicators and further examined how these interaction strategies relate to changes in psychological and functional outcomes. Moreover, the present findings reveal additional strategic interaction patterns that have received less attention in prior research, such as function-report-oriented interactions and passive content consumption patterns. These results suggest that care robot interactions cannot be characterized by a single mode of use; instead, multiple forms of engagement coexist and are associated with different outcome patterns.

First, Emotional Outpourers and Relational Connectors, both characterized by emotionally oriented interaction, showed significant decreases in depression and loneliness indicators. However, the two clusters differed in how emotional goals were expressed through interaction strategies. Emotional Outpourers demonstrated an emotional expression strategy centered on directly verbalizing emotions such as gratitude, affection, and apology. In contrast, Relational Connectors adopted a relational exchange strategy, organizing interactions around reciprocal exchanges such as greetings, social participation, and suggestions for shared activities. In other words, while Emotional Outpourers pursued emotional goals by directly articulating internal feelings, Relational Connectors expressed emotional meaning through reciprocal relational exchanges. This distinction suggests that similar emotional goals can be realized through different interaction structures. Nonetheless, both clusters shared the common feature of organizing interactions as emotionally meaningful experiences. These findings are consistent with Socioemotional Selectivity Theory (SST), which posits that older adults increasingly prioritize emotionally meaningful interactions as they age (Carstensen et al., 1999; Carstensen & DeLiema, 2018) and suggest that emotionally meaningful interactions may be associated with favorable psychological changes.

In contrast, Routine Anchors demonstrated a functional interaction strategy centered on daily management. In this cluster, functional utterances such as activity reports, meal-related conversations, and medication management accounted for a relatively high proportion of interactions. A particularly notable characteristic of this cluster was the repeated occurrence of report-type utterances, in which users conveyed their activities or physical states to the robot. On the surface, users in this cluster also maintained continuous conversations with the robot, and their interaction frequency did not appear particularly low. However, these interactions were closer to functional reporting rather than emotionally oriented communication or relationship building. This pattern contrasts sharply with the emotionally oriented clusters (Emotional Outpourers and Relational Connectors). In other words, even when users engage in similar levels of interaction with the robot, psychological outcomes may differ depending on whether those interactions are structured around emotional exchanges or functional reporting. The limited psychological change observed in Routine Anchors may therefore be better interpreted in relation to the functional nature of the interactions, rather than interaction frequency alone. These findings indicate that observed outcome patterns were more closely aligned with the purpose and emotional content of interactions than with interaction frequency alone.

Another contrast emerged in terms of the level of interaction engagement. Relational Connectors (Cluster 4) actively engaged in reciprocal interactions through greetings, social participation, and shared activities. In contrast, Entertainment Seekers repeatedly used content-based utterances such as requests for music playback or storytelling, reflecting a more passive interaction pattern. The analysis showed that loneliness indicators decreased to some extent in this cluster, but functional status tended to worsen. In particular, changes in WHODAS scores suggest that functional difficulties may have relatively increased in this group. One possible interpretation is that users experiencing physical discomfort or functional limitations may prefer low-effort interaction forms, such as passive content consumption, rather than active engagement. Activities such as listening to music or stories may provide a certain level of social stimulation and may be associated with reduced feelings of isolation. However, because such interactions do not require relational exchanges or active participation, they may be less closely aligned with deeper forms of emotional loneliness. This pattern may be interpreted as an example of the compensation strategy described in the Selection–Optimization–Compensation (SOC) model, suggesting that users with functional limitations may adopt interaction strategies that require less effort or cognitive demand.

Meanwhile, Adaptive Companions displayed a relatively balanced interaction structure between these more extreme strategies. Users in this cluster utilized entertainment requests, emotional expression, daily management, and information-seeking functions in a relatively balanced manner. Psychological indicators showed favorable pre–post changes, while functional status remained relatively stable. Rather than concentrating on a single interaction strategy, these users flexibly utilized multiple interaction resources depending on situational needs. This pattern can be interpreted in relation to the optimization strategy within the SOC framework (Baltes & Baltes, 1990). In other words, strategic flexibility in coordinating multiple interaction resources was observed alongside psychological and functional stability.

Taken together, the five clusters identified in this study indicate that interaction strategies in later life can be organized in different directions, including emotional, relational, functional, and entertainment-oriented patterns. These strategic differences can be understood as the result of the combined mechanisms described by Socioemotional Selectivity Theory (SST) and the Selection–Optimization–Compensation (SOC) framework. While emotionally meaningful goals tend to be prioritized in later life, the ways in which these goals are pursued may vary depending on individual resource conditions and strategic choices.

Importantly, observed outcome patterns differed not only by the amount or frequency of interaction but also by the strategic structure and purpose of interaction. Clusters centered on emotional expression or relational exchange showed more favorable psychological changes, whereas function-report-oriented or passive content-consumption clusters showed more limited or mixed changes. These findings suggest that care robot design may benefit from considering not only how often older adults interact with the robot but also whether interactions support emotionally meaningful and relational engagement.

Conclusion

This study examined long-term conversational data from older adults interacting with a socially assistive care robot in real-life environments. Using conversation-topic composition as a behavioral indicator, the

study categorized robot interaction behaviors and examined how different interaction profiles were associated with psychological, socio-emotional, and functional changes. The findings indicate that older adults used the same care robot in different ways and for different purposes, forming distinct behavioral interaction profiles.

In particular, the five behavioral clusters represented structural types reflecting distinct interaction strategies. Clusters characterized by emotional expression and relational exchange exhibited larger or more consistent pre–post decreases in depression and loneliness indicators, whereas function-report-oriented or passive content-consumption clusters showed more limited psychological changes or tendencies toward functional decline. These findings suggest that observed outcome patterns were not aligned simply with usage frequency or interaction quantity, but with how interactions were organized around specific goals and strategic structures.

From a theoretical perspective, this study contributes by linking Socioemotional Selectivity Theory (SST), which explains the prioritization of emotionally meaningful goals in later life, with the Selection–Optimization–Compensation (SOC) model, which explains adaptive strategies under resource constraints. The findings suggest that later-life technology interactions can be understood not only as functional use but also as strategic adaptation shaped by emotional goals and resource conditions.

From a practical perspective, the results imply that personalization strategies for care robot–based services should consider interaction profiles as well as functional preferences or usage frequency. For users characterized by emotional expression, designs emphasizing empathy and affective responsiveness may be appropriate, while relationally oriented users may require features that support reciprocal dialogue and social participation. For users characterized by functional reporting or passive content consumption, gradual prompts for broader participation and reciprocal interaction may be useful.

This study has several limitations. First, the analysis focused on a specific doll-type care robot within a particular national care service context, limiting generalizability. In addition, because low-interaction users were excluded to ensure stable estimation of topic distributions, these theoretically important users may reflect non-engagement or interaction barriers; thus, the cluster-based findings should be generalized primarily to older adults who engaged sufficiently with the robot's conversational function. Second, although generative AI–based classification was used, subtle emotional nuances or contextual meanings may not have been fully captured. Third, because the study relied on observational data, the observed pre–post changes should not be interpreted as causal effects of the robot or interaction profiles. Finally, because the clusters were based on aggregated topic distributions across the observation period rather than within-user temporal trajectories, they should be interpreted as dominant period-level interaction profiles rather than evidence of time-invariant conversational patterns; future research should examine topic shifts across early, middle, and later stages of use.

Future research should examine the generalizability of behavioral interaction strategies across different types of care robots and cultural contexts. In addition, further studies are needed to develop adaptive interaction systems capable of analyzing conversational patterns in real time, identifying users' interaction strategies, and dynamically adjusting responses accordingly. Such work would contribute to the development of more refined personalization strategies in care technologies and support the design of older adult–centered intelligent care systems.

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